

KNIGHT FLIGHT

IN 1921, WITH FUNDING for US airmail under threat, the Post Office decided to stage a daring night-flight experiment that would dramatically demonstrate the viability of a fully airborne transcontinental mail service. On February 22, two DH-4s took off from New York with mail bound for San Francisco, while another two set off from San Francisco with mail for New York.

They were to cover the stretch between Cheyenne and Chicago in darkness. Disaster soon struck. One of the eastbound pilots crashed in Nevada and was killed. On the westbound journey, only one load of mail got as far as Chicago, where the onward flight was cancelled as a snowstorm set in. In the middle of the night the only mail still in transit was passed to pilot

Jack Knight, who was to fly the segment from North Platte, Nebraska, to Omaha. Navigating by dead reckoning, and guided by bonfires that enthusiastic citizens on the ground below had lit along the way, Knight reached Omaha at 1:00am, only to find that the next pilot had failed to show up. So Knight had a strong coffee, tore a road-map of the onward route off a wall, and set off for Chicago, 435 miles (725km) away.

Exhausted and numb with cold in his open cockpit, Knight reached Iowa City at 5:00am and, after a refueling stop, flew on through snow to Maywood Airport, Chicago, arriving at 8:40am on February 23 to a hero's welcome. The westbound mail eventually reached New York 33 hours 25 minutes after leaving San Francisco.

TESTING, TESTING

Jack Knight, hero of the 1921 night-flight experiment, wears a radio microphone and helmet equipped with headphones to test radio air-to-ground communications.



The US Air Mail pilots were necessarily a tough bunch of individuals. Armed with pistols to protect the mail, they attempted to maintain regular flights over mountains and deserts in inadequate aircraft with only primitive instruments. Without aerial maps or reliable compasses, the pilots found their way either by following railroad tracks or looking out for landmarks that they recorded in a notebook – a church steeple, the orange roof of a barn, a water tower. When cloud cover was low they had little choice but to fly underneath it, running the risk of crashing into a hillside or even tall trees. If they encountered fog or a storm – a frequent occurrence – they had no choice but to land wherever they could. Even in flat, open terrain, a forced landing could be hazardous, as on the occasion succinctly reported by pilot Dean C. Smith: “Dead-sticked – flying low – only place available – on cow – killed cow – scared me.” The pilot would head off in search of

a farmstead, which might have a telephone, or a railroad, where he could flag down a passing train. Death was an accepted risk of the job. One pilot later recalled, “I would have been frightened if I had thought I would get maimed or crippled for life, but there was little chance of that. A mail pilot was usually killed outright.”

Initially some senior officials in the US Post Office were inclined to dismiss airmail as a passing fad. As the aircraft could only fly by day, they had to work in conjunction with trains. They were loaded up with mail from the nearest railroad in the morning, flew it some way along the route, and then handed it back to the trains for transport during the night. Thus most mail crossing the United States made, at best, only a small part of its journey by air, achieving an almost negligible improvement in delivery time at very considerable expense to the Post Office and risk to its pilots.

The only way to make the service viable was to establish night flying as normal practice, so that mail could travel from coast to coast entirely by air. During the war, pilots had repeatedly shown that flying at night was feasible, but a regular and reliable scheduled night operation over long distances would require something more than reliance on the skills and instincts of experienced pilots. The way forward was indicated by experiments conducted by the Army at McCook Field, near Dayton, Ohio, in 1923. They used rotating beacons and flashing markers to create a lighted airway between McCook Field and Columbus – a distance of over 70 miles (112km), along which Army pilots proved capable of making safe and regular night flights. The Post Office set out to translate this experiment into a continent-wide system of lighted airways. Night-time airmail services on the first illuminated stretch, from Chicago to Cheyenne, Wyoming, began in July 1924.

